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ChE 210 — Exam #1 — 11 October 2019

(1) (25 pts.) Consider a mixture of gasses consisting of three species, namely, A, B, and C. It is known that the mixture is made up of 40 mol% A, 18.75 wt % of B, 20 mol% of C. The molecular weight of A is 40kg/kmole and of C is 50 kg.kmole.

(a) Calculate the molecular weight of B in kg/kmol.

basis 100 kmols of mixture ⇒ 40 kmol A + 20 kmol C ⇒ kmol B = 100 − 60 = 40 kmol

$$\frac{18.75 \text{ kg B}}{\text{kg of mixture}} \times \frac{1}{M_B} = 40 \text{ kmol of B}$$

$$\Rightarrow M_B = \frac{18.75 \text{ kg B}}{40 \text{ kmol of B}} = 46.88 \frac{\text{kg B}}{\text{kmol B}}$$

you have 100 kmol basis and 100 kg basis Cannot use both

1	12
2	29
3	0
T	41

$M_B =$ ~~46.88~~ kg/kmol

(b) Calculate the mole average molecular weight of the mixture kg/kmole.

$$(0.4 \text{ A})\left(\frac{40 \text{ kg}}{\text{kmol}}\right) + 0.2 \times \frac{50 \text{ kg}}{\text{kmol}} + (0.4)(46.88 \text{ kg})$$

$$= 44.75 \frac{\text{kg}}{\text{kmol}} \text{ mixture}$$

OK

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$\bar{M} =$ 44.75 kg/kmol

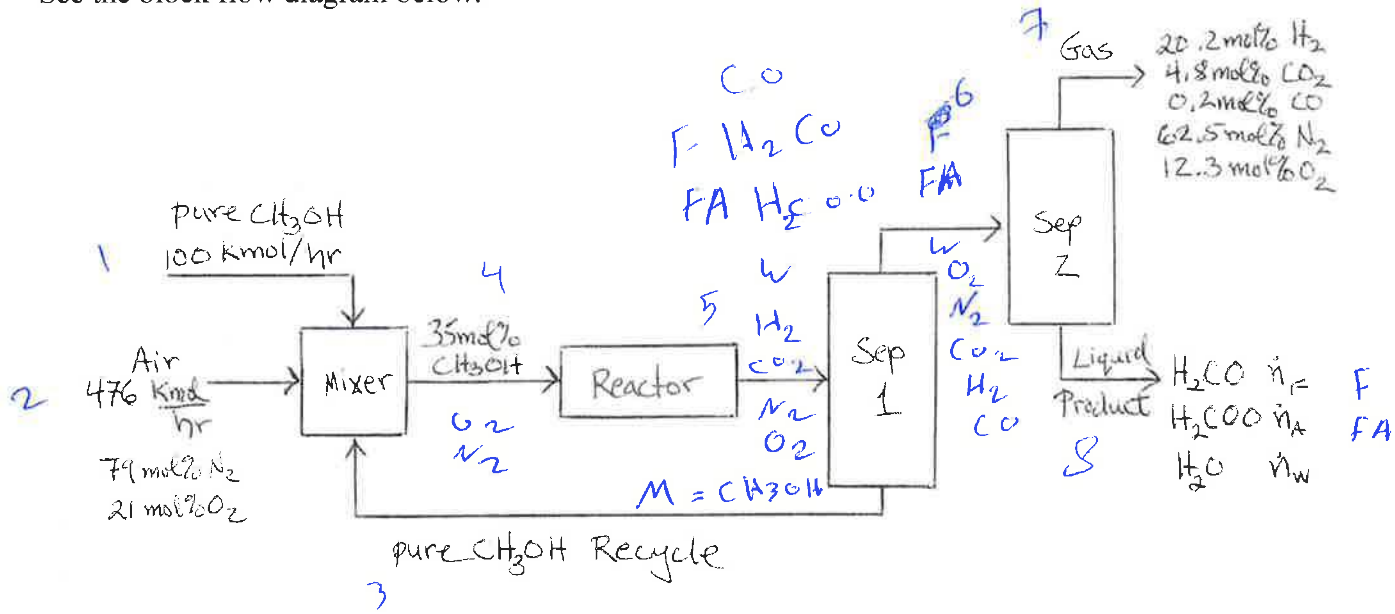
(2) (50 pts.) Formaldehyde (H_2CO) is produced by oxidation of methanol (CH_3OH). As shown, four reactions take place in the reactor. Pure methanol (CH_3OH) at 100 kmol/hr along with 476 kmol/hr of air (21 mol% O_2 and 79 mol% N_2) and a recycle stream of pure methanol are mixed and then enter the reactor. The reactor inlet stream is 35 mol% CH_3OH. The reactor exit stream goes to a separator with one exit stream of pure methanol that is recycled to the mixer and another stream that goes to a second separator that has an exit stream of gas with 20.2 mol% H_2, 4.8 mol % CO_2, 0.2 mol % CO, 62.5 mol% N_2, and 12.3 mol % O_2 and a liquid product stream containing formaldehyde (H_2CO), formic acid (H_2COO), and water (H_2O). See the block flow diagram below.

$$CH_3OH + \frac{1}{2}O_2 \rightarrow HCHO + H_2O \qquad (R1)$$

$$CH_3OH + O_2 \rightarrow HCOOH + H_2O \qquad (R2)$$

$$CH_3OH + O_2 \rightarrow CO + 2H_2O \qquad (R3)$$

$$CH_3OH + \frac{3}{2}O_2 \rightarrow CO_2 + 2H_2O \qquad (R4)$$



(a) Do a degree of freedom analysis following the method on page 105 of the textbook by Murphy. Fill in the table below giving a brief explanation for your entries. Compute the DOF. (10 pts.)

	Number	Explanation
4a. Stream Variables	~~38~~ 32	# of components in each stream
4b. System Variable	4	# of reactions
Total Independent Variables	~~32~~ 36	Sum of 4a + 4b
5a. Specified Flows	2	Know feedrates M + air
5b. Specified Stream Compositions	6	~~N2,~~ M known composition (N-1)
5c. Specified System Performances	0	No special constraints
5d. Material Balance Equations	~~30~~ 29	4 mixer + 9 reactor + 9 sep 1 + 8 sep 2
Total Independent Equations	38	
DOF	-2	

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C, O, H, N,

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PROBLEM 2 CONTINUED

(b) Calculate the product stream molar flow rates in kmol/hr for the formaldehyde (H_2CO) $\dot{n}_F$, formic acid (H_2COO) $\dot{n}_A$, and water (H_2O) $\dot{n}_W$ in kmol/hr. *Hint: Consider an atomic balance.* (30 pts.)

mixer: C: ~~100 kmol/h~~ Accum = in - out + gen - con no rxs

$\Rightarrow 100\ kmol/h + \dot{n}_{M,3} - 0.35\ \dot{n}_{M,4} = 0 \Rightarrow \dot{n}_{M,3} - 0.35\ \dot{n}_{M,4} = 100\ kmol/h$

H: $(4H)(100\ kmol/h) + (4H)(\dot{n}_{M,3}) - (0.35)(4H)\dot{n}_{M,4} = 0 \Rightarrow 4\dot{n}_{M,3} - 1.4\dot{n}_{M,4} = 400\ kmol/h$

O: $(476\ kmol/h)(2O)(0.21) + 100\ kmol - 0.35\dot{n}_4 - 2x_{4,O}\dot{n}_4 = 0$ (2 atom)

~~$\Rightarrow \dot{n}_{M,4} = 299.9\ kmol/h \Rightarrow \dot{n}_{M,3} = 204.972$~~

N_2: $0.79 \cdot (2N)(476\ kmol/h) = (1 - 0.35 - x_{4,O})\dot{n}_4$

Reactor: C: $0.35\dot{n}_4 - \dot{n}_{M,5} - \dot{n}_{F,5} - \dot{n}_{FA,5} - \dot{n}_{CO_2,5} - \dot{n}_{CO,5} = 0$

H: $4(0.35\dot{n}_4) - 2\dot{n}_{F,5} - 2\dot{n}_{FA,5} - 2\dot{n}_{H_2,5} - 4\dot{n}_{M,5} - 2\dot{n}_{W,5} = 0$

O: $2x_{O,4}\dot{n}_4 + 0.35\dot{n}_4 - \dot{n}_{CO,5} - \dot{n}_{F,5} - 2\dot{n}_{FA,5} - 2\dot{n}_{CO_2,5} - \dot{n}_{M,5} - \dot{n}_{W,5} = 0$

N_2: $(1 - 0.35 - x_{O,4})\dot{n}_4 = \dot{n}_{5,N_2}$

sep1: C: $\dot{n}_{5,CO} + \dot{n}_{F,5} + \dot{n}_{FA,5} + \dot{n}_{CO_2,5} + \dot{n}_{M,5} - \dot{n}_{F,6} - \dot{n}_{FA,6} - \dot{n}_{CO_2,6} - \dot{n}_{CO,6} - \dot{n}_{M,3} = 0$

H: $2\dot{n}_{F,5} + 2\dot{n}_{FA,5} + 2\dot{n}_{W,5} + 2\dot{n}_{H_2,5} - 2\dot{n}_{6,F} - 2\dot{n}_{6,FA} - 2\dot{n}_{H_2,6} - 2\dot{n}_{W,6} - 4\dot{n}_{3,M} = 0$ ~~$(2)\dot{n}_M = x_{F,8}\dot{n}_8$~~

O: $\dot{n}_{5,CO} + \dot{n}_{5,F} + 2\dot{n}_{5,FA} + \dot{n}_{5,W} + 2\dot{n}_{5,CO_2} + 2\dot{n}_{5,O_2} + \dot{n}_{5,M} - \dot{n}_{F,6} - 2\dot{n}_{FA,6} - \dot{n}_{W,6} - 2\dot{n}_{O_2,6} - 2\dot{n}_{CO_2,6} - \dot{n}_{CO,6} - \dot{n}_{M,3} = 0$

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$\dot{n}_F$ = ________ kmol H_2CO /min, $\dot{n}_A$ = ________ kmol H_2COO /min, $\dot{n}_W$ = ________ kmol H_2O/min

(c) Calculate the fractional conversion of methanol for the reactor (the single-pass conversion). (10 pts.)

$$\text{fractional conversion} = \frac{\text{mol of Methanol consumed}}{\text{mol of Methanol fed}} = 1$$

4

$$\text{selectivity} = \frac{\text{mol of desired product F}}{\text{mol of LR consumed}}$$

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$f_{C,methanol}$ = 1   Selectivity = ________

(25 pts.) The Ostwald Process produces nitric acid (HNO_3) by the oxidation of ammonia (NH_3). The process consists of three reactions:

R1: Ammonia reacts with oxygen (O_2) to form nitric oxide (NO) and water (H_2O).

R2: NO reacts with O_2 to form nitrogen dioxide (NO_2).

R3: NO_2 is bubbled through water to produce HNO_3 and NO.

(a) Write a balance chemical reaction equation for each of these three reaction. Make all of the stoichiometric coefficients the smallest whole numbers possible.

(b) Use a generation-consumption analysis to synthesize a reaction pathway to make nitric acid from ammonia and oxygen based on the three reaction in part (a) with no net generation or consumption of NO or NO_2. Water is allowed as a byproduct.

	R1	R2	R3	Overall Rxn
NH_3				
O_2				
NO				
NO_2				
H_2O				
HNO_3				

Write the net chemical equation for the process.

(c) Calculate the grams of NH_3 and grams of O_2 required to make 100 grams of nitric acid. Atomic Weights: H = 1 g/mol, N = 14 g/mol, and O = 16 g/mol.

Mass of O_2 = ______________ g

Mass of NH_3 = ______________ g

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2) b: Sep$_2$: C: $\dot{n}_{F,6} + \dot{n}_{FA,6} + \dot{n}_{CO_2,6} + \dot{n}_{CO,6} - 0.4\,\dot{n}_{7} - 0.2\,\dot{n}_{7}$

$$- \dot{n}_{F,8} - \dot{n}_{FA,8} = 0$$

H: $2\dot{n}_{F,6} + 2\dot{n}_{FA,6} + 2\dot{n}_{w,6} + 2\dot{n}_{H_2,6} - (0.202)\,2\,\dot{n}_{7}$

$$- 2\dot{n}_{F,8} - 2\dot{n}_{FA,8} - 2\dot{n}_{w,8} = 0$$

Write a balance on Nitrogen! Atomic bals For Overall system!

you wrote atomic balances on the separate unit operations, but only the mixer and overall systems were specified.

~~MIX~~

$40\ \text{kmol A} \times 40\ \text{kg/kmol of A} = 1600\ \text{kg of A}$

$20\ \text{kmol of C} \times 50\ \text{kg/kmol C} = 1000\ \text{kg}$

$1600 + 1000 +$

$\bigcirc$

$18.75 \times \frac{1}{\text{MB}} = 10 \quad \cancel{\text{kmol}}$

$0 - 46.88 \times 40 = 18.75$

0.4